

ARDUINO MEGA 2560 PLUG-AND-PLAY SHIELD (REV 7.0)

REV 7.0 VERIFIED

Zero-Soldering Modular Connection PCB for PecoDropDesktopApp/Arduino/RVM Arduino/RVM Arduino.ino

Why Use This Shield Instead of Hand Soldering?

The PecoDrop RVM firmware utilizes **8 ultrasonic distance sensors (32 wires)**, **6 high-torque servos (18 wires)**, **1 inductive metal sensor (3 wires)**, **1 HX711 24-bit load cell (4 wires)**, and **dual power buses**. Hand-soldering over 55 loose wires creates a fragile rat's nest susceptible to vibration disconnects, sensor crosstalk, and servo brown-out resets.

The Solution: This custom shield plugs directly on top of the Arduino Mega 2560. All sensors and servos plug into labeled 3-pin and 4-pin headers. **Zero loose jumper wires • Zero soldering during machine assembly • 100% plug-and-play maintenance.**



Figure 1: Custom Arduino Mega 2560 Expansion Shield REV 7.0 Plug-and-Play Hardware Map

Dual Isolated Power Architecture

- **12V DC Main Input:** Powers industrial NPN proximity sensor and on-board buck step-downs via 5.08mm screw terminal.
- **6V 5A High-Torque Servo Rail:** Dedicated copper traces rated for 5A continuous draw to prevent Arduino Mega brown-outs.
- **5V Logic Rail:** Clean, low-noise supply for all 8 HC-SR04 sensors and HX711 24-bit strain gauge ADC.

Ordering from JLCPCB / PCBWay (2-Minute Process)

- **Step 1:** Download [RVM_Arduino_Mega_PlugAndPlay_Shield_Gerbers.zip](#).
- **Step 2:** Visit [JLCPCB.com](#) or [PCBWay.com](#) and click *Add Gerber*.
- **Step 3:** Select **2 Layers**, Dimensions **101.6 × 53.3 mm**, 1.6mm FR-4, 1oz Copper.
- **Cost:** Approx **\$2.00 to \$4.00** for 5 boards with professional silkscreen and pre-drilled holes.

COMPLETE SENSOR & SERVO SOCKET CONNECTION DIRECTORY

All 3 Chambers

Connector ID	Connected Peripheral	Arduino Mega Pin	Connector Type	Pinout (Left to Right)
CH1_ENTR	Plastic Entrance Sonar	Trig: D9 • Echo: D10	4-Pin Dupont / JST-XH	[5V, D9, D10, GND]
CH1_BOT	Plastic Sizing Bottom	Trig: D22 • Echo: D23	4-Pin Dupont / JST-XH	[5V, D22, D23, GND]
CH1_MID	Plastic Sizing Middle	Trig: D24 • Echo: D41	4-Pin Dupont / JST-XH	[5V, D24, D41, GND]
CH1_TOP	Plastic Sizing Top	Trig: D42 • Echo: D43	4-Pin Dupont / JST-XH	[5V, D42, D43, GND]
CH1_IRIS	Plastic Upper Iris Servo	PWM Pin 11	3-Pin Standard Servo	[D11, 6V_SERVO, GND]
CH1_DROP	Plastic Drop Gate Servo	PWM Pin 12	3-Pin Standard Servo	[D12, 6V_SERVO, GND]
CH2_ENTR	Metal Can Entrance Sonar	Trig: D25 • Echo: D26	4-Pin Dupont / JST-XH	[5V, D25, D26, GND]
CH2_BOT	Can Sizing Bottom	Trig: D29 • Echo: D30	4-Pin Dupont / JST-XH	[5V, D29, D30, GND]
CH2_MID	Can Sizing Middle	Trig: D31 • Echo: D44	4-Pin Dupont / JST-XH	[5V, D31, D44, GND]
CH2_TOP	Can Sizing Top	Trig: D45 • Echo: D46	4-Pin Dupont / JST-XH	[5V, D45, D46, GND]
CH2_IND	Inductive Proximity Sensor	Signal: Pin 32	3-Pin Screw / JST	[12V_RAW, D32, GND]
CH2_IRIS	Can Upper Iris Servo	PWM Pin 27	3-Pin Standard Servo	[D27, 6V_SERVO, GND]
CH2_DROP	Can Drop Gate Servo	PWM Pin 28	3-Pin Standard Servo	[D28, 6V_SERVO, GND]
CH3_ENTR	Paper Top Entrance Sonar	Trig: D33 • Echo: D34	4-Pin Dupont / JST-XH	[5V, D33, D34, GND]
CH3_BOT	Paper Bottom Arrival Sonar	Trig: D39 • Echo: D40	4-Pin Dupont / JST-XH	[5V, D39, D40, GND]
CH3_HX	HX711 24-Bit Load Cell	DOUT: D37 • SCK: D38	4-Pin Dupont / JST-XH	[5V, D37, D38, GND]
CH3_IRIS	Paper Upper Iris Servo	PWM Pin 35	3-Pin Standard Servo	[D35, 6V_SERVO, GND]
CH3_DROP	Paper Drop Gate Servo	PWM Pin 36	3-Pin Standard Servo	[D36, 6V_SERVO, GND]
TB_PWR_12V	Main 12V 10A DC Input	VIN / RAW	5.08mm Screw Terminal	[GND, +12V_RAW]
TB_PWR_6VS	Dedicated 6V 5A Servo Input	6V SERVO BUS	5.08mm Screw Terminal	[GND, +6V_SERVO]

Urdu Summary (سولڈرنگ سے نجات اور پلگ اینڈ پلے کنکشن گائیڈ)

یہ کسٹم شیلڈ آرڈوینو میگا ۲۵۶۰ پر براہ راست فٹ ہو جاتی ہے۔ تمام الٹراسونک سنسز، سروو موٹرز، میٹل سنسر اور لوڈ سیل کے لیے الگ الگ ۳ اور ۴ پین کے کنیکٹرز فراہم کیے گئے ہیں۔ آپ کو مشین کی اسمبلنگ کے دوران ایک بھی تار سولڈر نہیں کرنی پڑے گی۔ ہر کیبل اپنے مخصوص کنیکٹر میں براہ راست پلگ ہو جاتی ہے جس سے وقت اور تقاض کی مکمل بچت ہوتی ہے۔

PRODUCTION BILL OF MATERIALS (BOM)

Complete Component Procurement List

Item	Component / Description	Package / Pitch	Quantity	Recommended MPN / Part	Function / Location
1	Arduino Mega Header Kit	2.54mm Stackable	1 Set	Generic Mega 2560 Kit	Plugs onto Arduino Mega 2560
2	2-Pin Screw Terminal Block	5.08mm Through-hole	2 Pcs	KF301-2P / Phoenix 1715721	12V Main & 6V Servo Power Input
3	4-Pin Right Angle Header / JST	2.54mm Pitch	10 Pcs	JST-XH-4P / 2.54mm Pin Header	All 8 HC-SR04 & HX711 Sockets
4	3-Pin Right Angle Header / JST	2.54mm Pitch	7 Pcs	Standard 3-Pin Servo Header	6x MG996R Servos & Inductive Sensor
5	Low-ESR Electrolytic Capacitor	Radial Ø10mm	2 Pcs	Nichicon 1000µF 25V (UHW1E102MPD)	Power decoupling & servo buffer
6	Multilayer Ceramic Capacitor (MLCC)	0.1µF 50V (C1206 / Radial)	6 Pcs	Yageo CC1206KKX7R9BB104	High-frequency noise decoupling
7	Optocoupler Photocoupler	DIP-4 / SMD	1 Pc	Everlight EL817C / Sharp PC817	12V Inductive Sensor isolation
8	Flyback Protection Diode	DO-41 Through-hole	6 Pcs	Vishay 1N4007-E3/54	Servo inductive spike clamping
9	Pull-up Resistor 10kΩ 1/4W	Through-hole Axial / 0805	2 Pcs	Yageo CFR-25JR-52-10K	Inductive & Opto pull-up bias
10	TVS Surge Suppression Diode	DO-214AA (SMB)	2 Pcs	Littelfuse SMBJ15A	Transient spike protection on VIN

Home Fabrication via Toner Transfer / CNC

If fabricating locally instead of ordering from JLCPCB:

- Open [RVM_Arduino_Mega_Shield_1to1_Drill_and_Etch_Film.pdf](#).
- Print Page 2 (Mirrored Mask) on a laser printer at **100% scale** onto glossy transfer paper.
- Verify the **100.00 mm calibration ruler** with digital calipers.
- Iron onto 1.6mm single/double-sided FR-4 copper board at 200°C for 3 minutes, then etch in Ferric Chloride.

Initial Power-Up Checklist

- Step 1:** Before plugging the shield onto the Arduino Mega, connect 12V and verify 5.0V on logic pins and 6.0V on servo headers using a multimeter.
- Step 2:** Plug shield onto the Arduino Mega 2560 gently, ensuring all pin headers seat without bending.
- Step 3:** Plug in each sensor cable one by one. Run `CMD:TEST_SERVOS` and `CMD:TEST_SENSORS` over Serial Monitor at 115200 baud.